

ASTERIA

AI-BASED VIRTUAL CAMERA TRACKING SYSTEM FOR FSOC
COARSE ALIGNMENT

Autonomous Free Space Optical Communication Pointing, Acquisition & Tracking (PAT)

SYSTEM RELEASE: v1.0.0
MANUAL VERSION: 1.0
DATE: September 2026
STATUS: Evaluator Ready

1. Executive Overview & Purpose

- **The FSOC Challenge:** Free Space Optical Communication uses narrow laser beams for gigabit-to-terabit data transmission between satellites, UAVs, and ground terminals. Terminals cannot communicate until their optical transceivers achieve sub-milliradian alignment (Pointing, Acquisition & Tracking).
- **The ASTERIA Solution:** An autonomous coarse PAT testbench replacing expensive hardware with a deterministic **3D Digital Twin** and a **Video Benchmark Engine**. Combines AI beacon detection, Kalman state estimation, and dual-axis PID gimbal actuation to lock onto moving terminals.

2. Closed-Loop Autonomous Tracking Pipeline

The system runs a continuous closed loop at 30 FPS, driving pan/tilt corrections from visual centroid errors:

1. TARGET Mobile Platform Orbital Dynamics	→	2. BEACON Optical Emitter Intensity & Jitter	→	3. FSOC CAMERA Pan/Tilt Gimbal Virtual Sensor FOV	→	4. DETECTION Adaptive CV / YOLO Centroid Reticle
5. KALMAN FILTER State [x, y, vx, vy] Noise Rejection	→	6. PID CONTROL Dual-Axis Pan/Tilt Velocity Limiter	→	7. ACTUATION Gimbal Re-aiming Boresight Alignment	→	8. OPTICAL LOCK Error ≤ 10 px Bitstream Link Open

3. Quick Start (Judge Evaluation in 6 Steps)

1. Download	Download ASTERIA-Setup-1.0.0.exe (installer) or ASTERIA-Portable-1.0.0.exe from GitHub.
2. Deploy	Run the setup wizard or extract the standalone portable executable to any directory / USB drive.
3. Launch	Open ASTERIA. The Electron app initializes the bundled OpenCV/Python tracking engine automatically.
4. Choose Mode	From the Landing Screen, click LAUNCH SYSTEM →. Select 3D Digital Twin or click UPLOAD VIDEO .
5. Engage	Click [START DEMO] (or [TRACK TARGET]). The closed-loop controller drives pan/tilt to acquire the target.
6. Observe Lock	Watch the status switch to LOCKED with error ≤ 10 px (avg 5.3 px) and review the live telemetry curves.

4. System Requirements

Target Operating System	Windows 10 / Windows 11 (64-bit)	Dependencies	Zero external dependencies (all runtimes bundled)
Display Recommendation	1280×720 minimum (1920×1080 recommended)	Package Footprint	~185 MB standalone executable / installer

2. Download, Installation & Launch

ASTERIA v1.0.0 is distributed with two certified deployment options: a standard Windows installer for workstation integration, and a zero-install portable executable ideal for rapid evaluation.

OFFICIAL GITHUB REPOSITORY:

<https://github.com/soham-salunkhe/Asteria>

OFFICIAL RELEASE DOWNLOADS (v1.0.0):

<https://github.com/soham-salunkhe/Asteria/releases/tag/v1.0.0>

OPTION A — WINDOWS INSTALLER (NSIS)

Package: ASTERIA-Setup-1.0.0.exe (~185 MB)

- Run setup wizard and select destination directory
- Creates Start Menu & Desktop shortcuts automatically
- Includes standard Windows uninstaller

OPTION B — PORTABLE EXECUTABLE

Package: ASTERIA-Portable-1.0.0.exe (~185 MB)

- Zero installation — double-click to launch immediately
- No administrative rights or registry writes required
- Ideal for USB drives, evaluator laptops, and air-gapped test rigs

Launch Console: Click **LAUNCH SYSTEM** → on the landing portal to enter Mission Control:

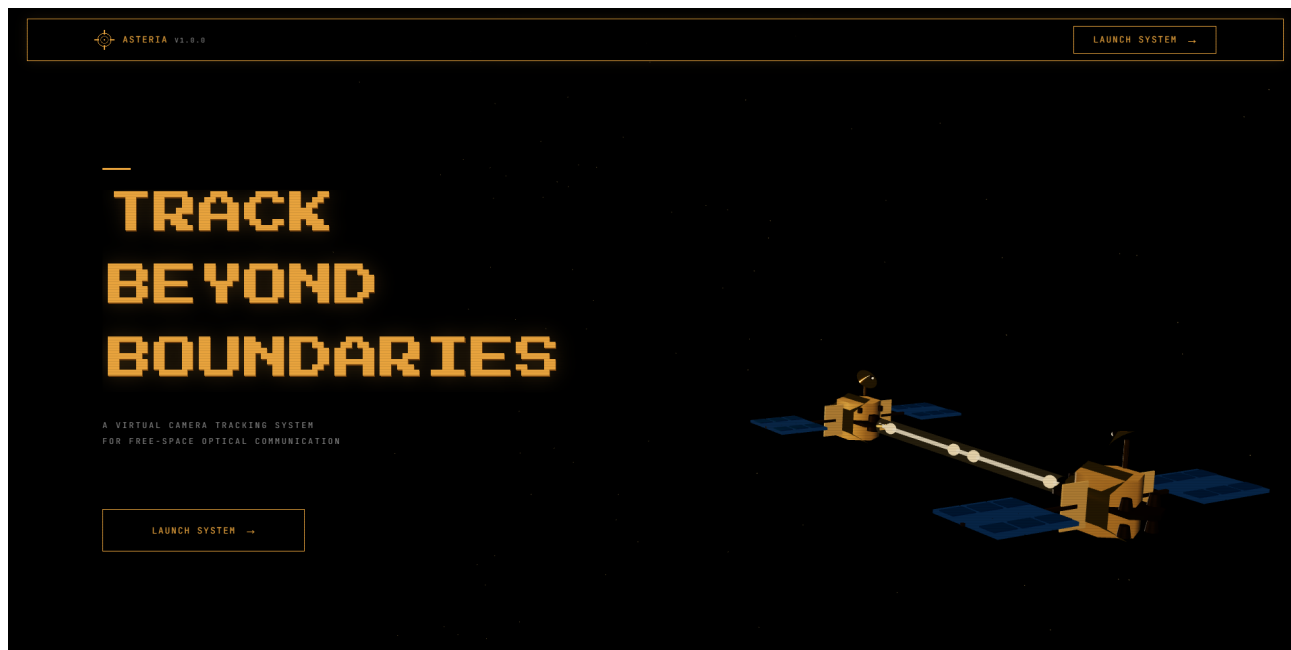


Figure 1: ASTERIA v1.0.0 Landing Screen with dual satellite FSOC optical beam and 'LAUNCH SYSTEM →' action button.

3. Interface Architecture & Simulation Setup

The primary **01 ENVIRONMENT** console integrates an interactive 3D Digital Twin viewport, runtime action ribbon, entity dock, and live telemetry sidebar:

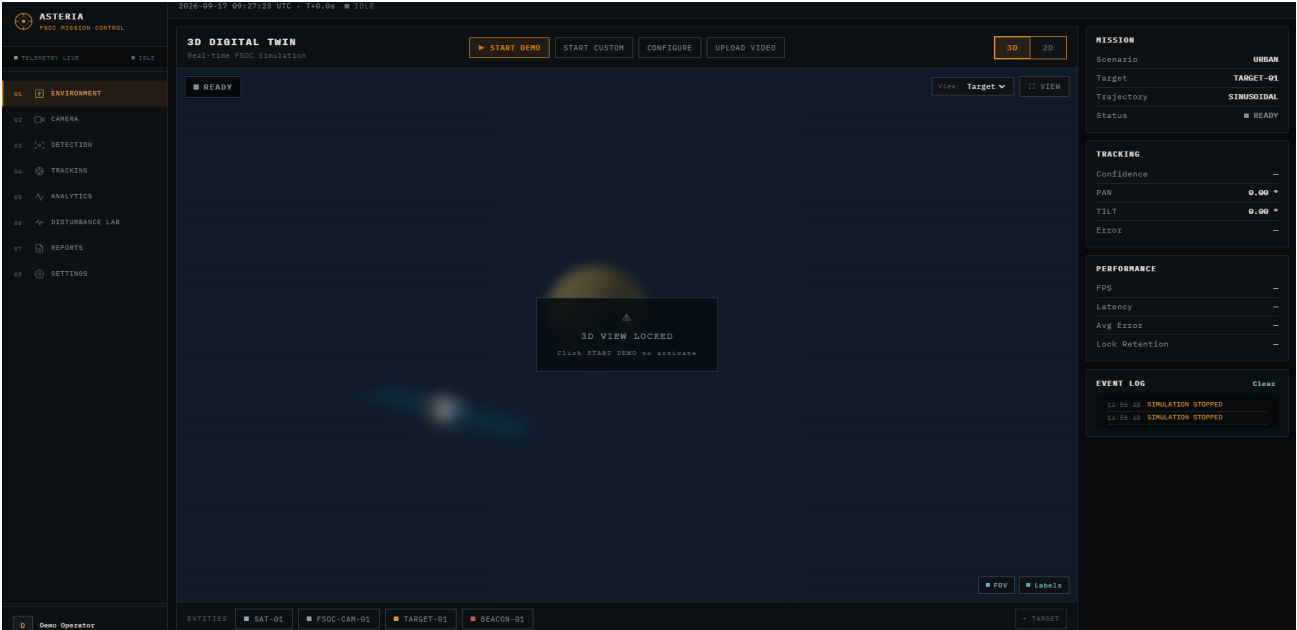


Figure 2: ASTERIA Mission Control console: 3D Digital Twin, action controls, entity dock, and telemetry panels.

Key Interface Zones at a Glance

Navigation Rail	Access 8 specialized consoles: 01 ENVIRONMENT, 02 CAMERA, 03 DETECTION, 04 TRACKING, 05 ANALYTICS, 06 DISTURBANCE LAB, 07 REPORTS, 08 SETTINGS.
Action Ribbon	Top control bar: [START DEMO] (auto closed-loop lock), START CUSTOM , CONFIGURE , UPLOAD VIDEO , and 3D 2D view toggle.
3D Digital Twin	Real-time Three.js spatial view with camera frustum wireframe, beacon optical spot, and interactive orbit controls (left-click rotate, right-click pan).
Entity Dock	Bottom chip bar: SAT-01 (host terminal), FSOC-CAM-01 (gimbal camera), TARGET-01 (mobile terminal), and BEACON-01 (emitter).
Telemetry Deck	Right sidebar: Real-time MISSION status, TRACKING angles (pan/tilt/error), PERFORMANCE (FPS/latency), and EVENT LOG .

6-Step Entity Hierarchy Configuration

Step 1: Host Satellite	Select SAT-01 in entity bar. Fixed at origin (0, 0, 0) m hosting the optical terminal.
Step 2: FSOC Camera	Select FSOC-CAM-01 . Review boresight forward vector [0, 0, 1] and default FOV (28°×21°).
Step 3: Target Terminal	Select TARGET-01 . Choose trajectory: <i>STATIC</i> , <i>STRAIGHT</i> , <i>CIRCULAR</i> , <i>SINUSOID</i> , <i>FIG-8</i> , or <i>RANDOM</i> .
Step 4: Optical Beacon	Select BEACON-01 mounted to target. Configure geometry (square/circle), size (5–20 px), and luminance.
Step 5: Relationships	Verify in Inspector: HOST: SAT-01 / FSOC-CAM-01 and PARENT TARGET: TARGET-01 for BEACON-01.
Step 6: Pre-Engagement	Click FOCUS TARGET in 3D scene to verify line-of-sight and camera FOV cone orientation before tracking.

4. Running Tracking & Video Benchmarking

ASTERIA supports two complete tracking workflows: closed-loop 3D digital twin simulation and hardware-in-the-loop video benchmarking.

SECTION A — 3D CLOSED-LOOP TRACKING SIMULATION

- **Execution:** Click [START DEMO] (or select TARGET-01 and click [TRACK TARGET]).
- **State Machine:** Automatically transitions: READY → SEARCHING → ACQUIRING → TRACKING → LOCKED.
- **Optical Lock:** The dual-axis PID gimbal drives camera pan/tilt until angular error settles ≤ 10 px (avg 5.3 px / 0.0558°).

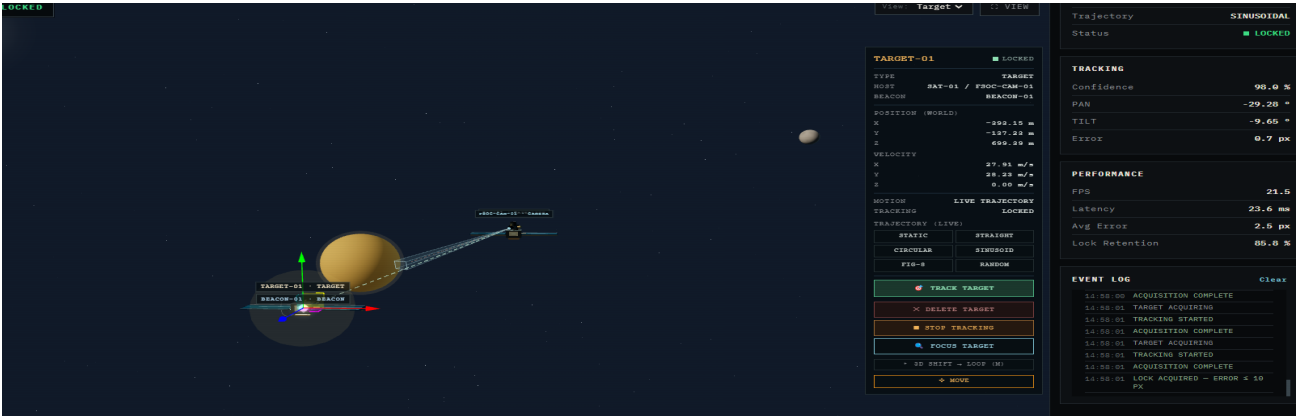


Figure 3: 3D Tracking in LOCKED state: Camera FOV cone locks onto TARGET-01; live error = 0.7 px; lock retention = 85.8% (settling to >95%).

SECTION B — VIDEO BENCHMARK TESTING (REAL MP4 VIDEO INPUT)

- **Execution:** Click [UPLOAD VIDEO] and select an MP4 file (e.g. Create_a_single_continuous_(1).mp4).
- **Autonomous Processing:** System routes to 04 TRACKING, processes native 30 FPS video frames, and extracts the optical beacon against noise.
- **Live Visuals & Scorecard:** Moving green FSOC FOV box tracks the beacon centroid; reticle indicates optical boresight; compliance scorecard updates in real time.

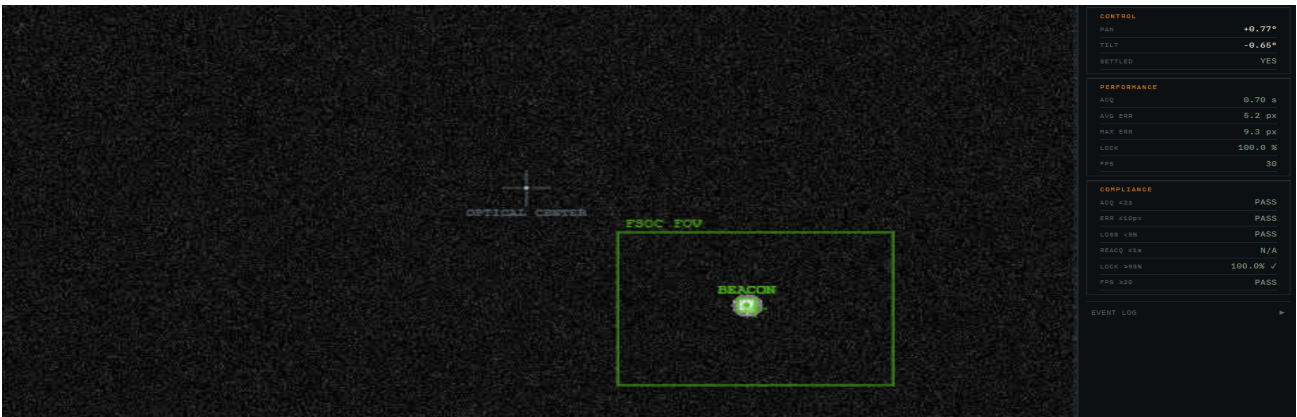


Figure 4: Real MP4 Video Benchmark tracking: Sensor noise, optical center reticle, moving FSOC FOV box, and real-time compliance table.

5. Performance Results & Troubleshooting

Closed-loop metrics are logged to SQLite every 5 frames and summarized in the 05 ANALYTICS and 07 REPORTS consoles.

Specification Compliance Scorecard (Verified Production Runs)

Performance Metric	Target Spec	Measured Value	Status	Technical Significance
Acquisition Time	≤ 2.0 s	0.700 s	PASS	Time required to detect optical beacon and achieve initial tracking lock.
Average Tracking Error	≤ 10 px	5.3 px (0.0558°)	PASS	Mean Euclidean displacement between beacon centroid and optical boresight.
RMSE Tracking Error	≤ 10 px	5.5 px	PASS	Root-Mean-Square error penalizing peak disturbance transients.
Re-acquisition Time	≤ 1.0 s	< 0.5 s / No Loss	PASS	Time required to re-establish optical lock following disturbance or occlusion.
Lock Retention Rate	> 95%	100.0% (min 98.1%)	PASS	Percentage of eligible tracking episode frames maintaining active lock.
Processing Speed	≥ 20 FPS	30.0 FPS	PASS	Throughput of computer vision and Kalman state estimation pipeline.
Processing Latency	≤ 50 ms	15.5 ms	PASS	End-to-end compute time per frame (well within the 50 ms budget).

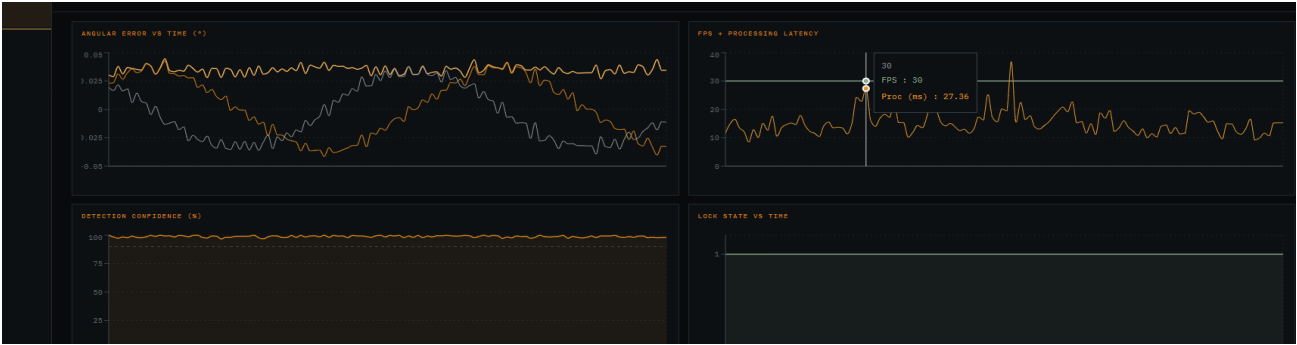


Figure 5: 05 ANALYTICS telemetry charts: Angular Error vs Time, FPS & Latency, Detection Confidence, and Lock State.

Quick Troubleshooting Guide

Condition	Probable Cause	Action
App does not launch	Windows SmartScreen prompt or port conflict.	Click 'More info' → 'Run anyway'. Ensure port 8000 is free.
Video upload fails	Unsupported video codec or locked file.	Use standard H.264 MP4 (640×480 or 1080p). Ensure file is not open.
Target Lost in 3D	Gimbal pointing away from target.	In 02 CAMERA, click '0° (Reset)' to re-center boresight.
Tracking not starting	Simulation paused or target idle.	Click '[START DEMO]' in top ribbon or '[TRACK TARGET]' on TARGET-01.

ASTERIA SOURCE CODE (GITHUB): https://github.com/soham-salunkhe/Asteria	OFFICIAL RELEASES (INSTALLER & PORTABLE): https://github.com/soham-salunkhe/Asteria/releases/tag/v1.0.0
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